Morley's Theorem

All Roads Lead to Geometry

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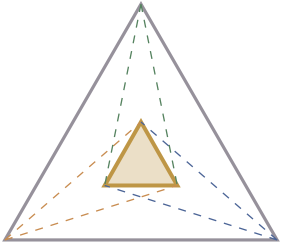
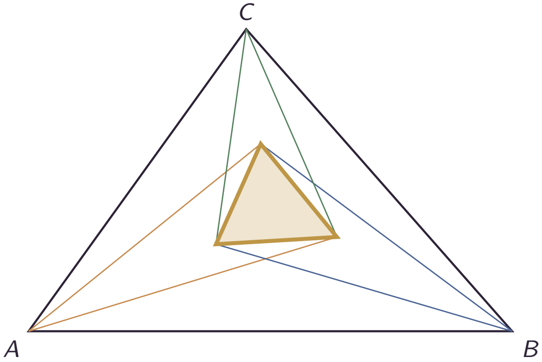


A map of our exploration

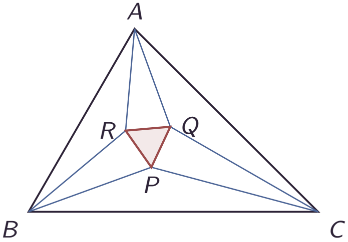
The Theorem

Morley's Trisector Theorem

The three points of intersection of the **adjacent trisectors** of the angles of any triangle form an **equilateral triangle**.



Proof 1: Trigonometric Proof (Part 1)



Law of Sines Computation

Let $\angle A = 3\alpha, \angle B = 3\beta, \angle C = 3\gamma$, so $\alpha + \beta + \gamma = 60^\circ$. Assume the circumradius of $\triangle ABC$ is 1. Then $BC = 2\sin(3\alpha)$.

In $\triangle BPC$, the Law of Sines gives:

$$\frac{BP}{\sin\gamma} = \frac{BC}{\sin(180^\circ - \beta - \gamma)} = \frac{2\sin(3\alpha)}{\sin(60^\circ - \alpha)}$$

The triple angle identity $\sin(3\alpha) = 4\sin\alpha\sin(60^\circ + \alpha)\sin(60^\circ - \alpha)$ cancels the denominator:

$$\begin{aligned} BP &= \frac{8\sin\alpha\sin(60^\circ + \alpha)\sin(60^\circ - \alpha)\sin\gamma}{\sin(60^\circ - \alpha)} \\ &= 8\sin\alpha\sin\gamma\sin(60^\circ + \alpha) \end{aligned}$$

By symmetry, in $\triangle ARB$, we find:

$$BR = 8\sin\gamma\sin\alpha\sin(60^\circ + \gamma)$$

Proof 1: Trigonometric Proof (Part 2)

Law of Cosines & Reference Triangle
In $\triangle BPR$, we apply the Law of Cosines:

$$PR^2 = BP^2 + BR^2 - 2(BP)(BR)\cos\beta$$

Substituting our expressions for BP and BR:

$$\begin{aligned} PR^2 = 64\sin^2\alpha\sin^2\gamma\big[&\sin^2(60^\circ+\alpha) \\ &+\sin^2(60^\circ+\gamma) \\ &-2\sin(60^\circ+\alpha)\sin(60^\circ+\gamma)\cos\beta\big] \end{aligned}$$

Key Step: Consider a reference triangle with angles $60^\circ+\alpha$, $60^\circ+\gamma$, and β (which sum to 180°). If its circumradius is 1, its sides are $2\sin(60^\circ+\alpha)$, $2\sin(60^\circ+\gamma)$, and $2\sin\beta$.
Applying the Law of Cosines to this reference triangle reveals that the bracketed term is exactly $\sin^2\beta$!

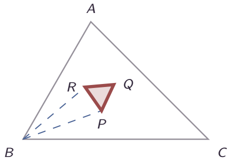
Thus, the expression collapses to:

$$PR^2 = 64\sin^2\alpha\sin^2\gamma\sin^2\beta$$

$$\Rightarrow PR = 8\sin\alpha\sin\beta\sin\gamma$$

This expression is symmetric in α, β, γ. By symmetry, $PQ = QR = 8\sin\alpha\sin\beta\sin\gamma$.

Therefore, $PR = PQ = QR$.
$\triangle PQR$ is equilateral!



Proof 2: Algebraic Proof (Part 1)

Turns on the Unit Circle

We represent points as *turns* (complex numbers with $|t| = 1$, so $t^* = 1/t$). Let $\omega = e^{2\pi i/3}$. A key property is that $\triangle pqr$ is equilateral if and only if:

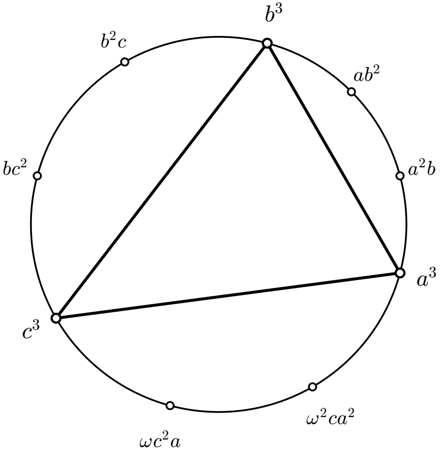
$$p + \omega q + \omega^2 r = 0 \quad \text{(or with } \omega \leftrightarrow \omega^2\text{)}$$

The Setup: Let the circumcircle of $\triangle ABC$ be the unit circle U. We assign the vertices A, B, C the affixes a^3, b^3, c^3.

The points of trisection of the arcs AB, BC, CA are elegantly given by the turn products:

$$a^2b, ab^2; \quad b^2c, bc^2; \quad \omega c^2a, \omega^2ca^2$$

The interior trisectors of the triangle correspond to chords connecting these specific points on U.



Proof 2: Algebraic Proof (Part 2)

Chord Intersections

The standard equation of a chord joining turns t, u is $z + tuz^* = t + u$. The Morley vertex x is the intersection of chords (ab^2, c^3) and $(b^3, \omega c^2 a)$. Solving the system:

$$x + ab^2c^3x^* = ab^2 + c^3$$
$$x + \omega ab^3c^2x^* = \omega ac^2 + b^3$$

Subtracting yields:

$$a^2b^2c^2x^* = \omega^2(ab^2 - a^2b) + ac^2 - \omega a^2c + \omega abc$$

By cyclically permuting a, b, c, we obtain similar expressions for y^* and z^*.

Symmetry & Conclusion

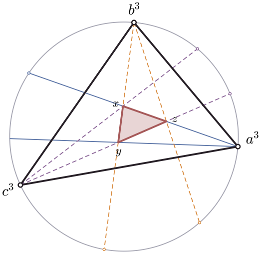
Summing the cyclic expressions:

$$x^* + \omega^2y^* + \omega z^* = 0$$

Taking the complex conjugate (since $\omega^* = \omega^2$):

$$x + \omega y + \omega^2 z = 0$$

This is exactly the condition for $\triangle xyz$ to be equilateral!



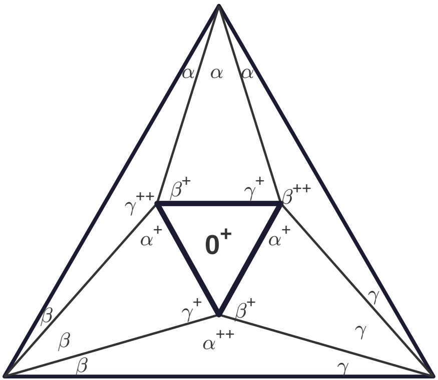
Proof 3: Synthetic Proof (Part 1)

Defining the Seven Pieces

Conway's proof reverses the theorem: start with an equilateral triangle and build $\triangle ABC$ from 7 pieces.
Let $\alpha = A/3, \beta = B/3, \gamma = C/3$, so $\alpha+\beta+\gamma = 60^\circ$. Define $\theta^* = \theta + 60^\circ$ and $\theta^{**} = \theta + 120^\circ$.
We define the following triangles (their angles sum to 180°):

- **1 Central Equilateral:** $0^*, 0^*, 0^*$. Scaled to side-length 1.
- **3 Abutting Triangles:** e.g., $\alpha, \beta^*, \gamma^*$. Scaled so the side opposite α is 1.
- **3 Spoke Triangles:** e.g., $\alpha^{**}, \beta, \gamma$. (Scaling defined on next slide).

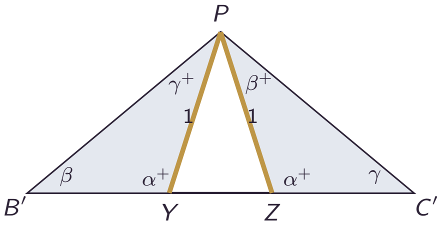
These 7 pieces are our building blocks.



Proof 3: Synthetic Proof (Part 2)

Sizing the Spoke Triangles

We must guarantee the pieces fit together. Consider the spoke triangle $PB'C'$ with angles α^{**} at P, β at B', and γ at C'.
Fix points Y and Z on the line $B'C'$ such that the angles $\angle PYB'$ and $\angle PZC'$ equal α^*.
Because $\angle B'PY = 180^\circ - \beta - \alpha^* = \gamma^*$, the sub-triangle $\triangle B'PY$ has angles $\beta, \alpha^*, \gamma^*$. This makes it strictly similar to our abutting triangle!
We scale the spoke triangle so that $PY = PZ = 1$. Now the side PY perfectly matches the side length 1 of the abutting triangle.



Proof 3: Synthetic Proof (Part 3)

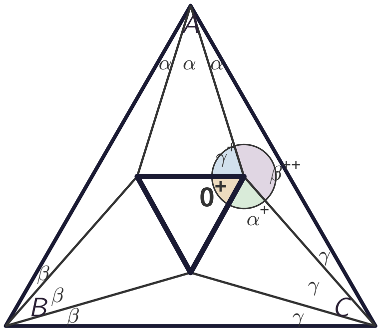
Assembling the Master Triangle

We attach the three abutting triangles to the central equilateral $\triangle PQR$. Does the spoke triangle fit into the remaining gap at the highlighted right vertex (Q)?

The angles meeting at Q from the equilateral and abutting triangles are 60° (or 0^+), α^+, and γ^+. Their sum is $60^\circ + (\alpha + 60^\circ) + (\gamma + 60^\circ) = 180^\circ + \alpha + \gamma$. Since $\alpha + \beta + \gamma = 60^\circ$, this is $240^\circ - \beta$. The remaining gap is $360^\circ - (240^\circ - \beta) = 120^\circ + \beta = \beta^{++}$. This exactly accommodates the spoke triangle $QA'C'$!

By ASA congruence ($\triangle C'QZ \cong \triangle CQP$), $C'Q = CQ$, so C' is precisely the vertex C.

Assembling all 7 pieces yields a perfect triangle ABC. The total angle at C is 3γ, proving the adjacent lines are trisectors!



Proof 4: The Group Theory Lens

The Affine Group Shift

Connes shifts the framework from Euclidean geometry to the affine group of the line over $\mathbb{C}$.

- **Transformations** take the form $g(x) = ax + b$, represented as matrices $g = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$.
- **Rotations:** Let g_1, g_2, g_3 be rotations around A, B, C. The angle of each is two-thirds of the full triangle angle.

Fixed Points as Intersections

For any affine map $g(x) = ax + b$ (with $a \neq 1$), the unique fixed point is:

$$\text{fix}(g) = \frac{b}{1 - a}$$

Connes proves that the fixed points of the pairwise products g_1g_2, g_2g_3, g_3g_1 are precisely the trisector intersections α, β, γ.

Proof 4: Equivalence (The Proof)

The Algebraic Definition of an Equilateral Triangle

Three points α, β, γ form an equilateral triangle iff $\alpha + j\beta + j^2\gamma = 0$ (where $j^3 = 1, j \neq 1$).

The Crucial Computation

Setting $g_1^3 g_2^3 g_3^3 = 1$ forces the translational component $b = 0$. Computing b explicitly using $a_1 a_2 a_3 = j$ yields the factorization:

The Translational Part b

$$b = -ja_1^2 a_2 (a_1 - j)(a_2 - j)(a_3 - j) \cdot (\alpha + j\beta + j^2\gamma)$$

Conclusion

By hypothesis the pairwise products are not pure translations, so each $(a_k - j) \neq 0$. Since $b = 0$ is a geometric certainty ($g_1^3 g_2^3 g_3^3 = 1$ for any triangle), the remaining factor must vanish:

$$\alpha + j\beta + j^2\gamma = 0$$

This is the equilateral triangle condition. No diagram required.

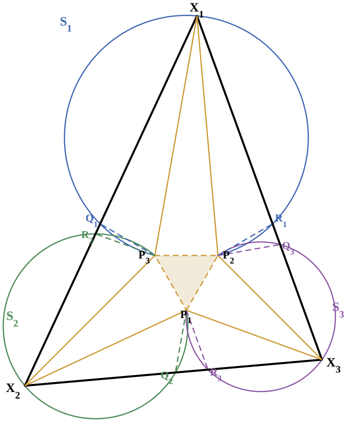
Proof 5: The Circle Intersection Lens (Part 1)

A New Reverse Construction

We start with an equilateral triangle $P_1P_2P_3$ and angles $\alpha_1, \alpha_2, \alpha_3$ (summing to $\pi/3$). We construct a triangle with angles $3\alpha_1, 3\alpha_2, 3\alpha_3$ whose trisectors meet at P_1, P_2, P_3.

The Circles and Chords

- Construct a circle S_1 through P_2, P_3 such that chord P_2P_3 subtends angle α_1.
- Find points Q_1, R_1 on S_1 such that chords Q_1P_3 and P_2R_1 equal P_2P_3.
- Repeat to construct circles S_2, S_3 and points Q_2, R_2, Q_3, R_3.
- Let the line Q_1R_2 and line R_1Q_3 intersect at X_1. Form X_2 and X_3 similarly.



Proof 5: The Circle Intersection Lens (Part 2)

The Pentagonal Angle Sum

By elementary angle chasing, $P_iQ_jX_i = P_iR_jX_k = \frac{2\pi}{3} - \alpha_i$. Summing the angles of the pentagon $P_2Q_1X_1R_1P_3$ (which must equal 3π), we deduce exactly that $\angle Q_1X_1R_1 = 3\alpha_1$.

Trisection on the Circle

- Because $\angle Q_1X_1R_1 = 3\alpha_1$ correctly subtends the arc on S_1, X_1 must lie exactly on the circle S_1.
- Because the chords Q_1P_3, P_3P_2, and P_2R_1 were specifically constructed to be equal in length, they must subtend equal angles from X_1.
- Thus, the lines X_1P_2 and X_1P_3 cleanly **trisect** the angle X_1.

Repeating this symmetry for all three vertices constructs a triangle $X_1X_2X_3$ with angles $3\alpha_i$ correctly trisected by P_1, P_2, P_3, proving Morley's Theorem!

Beyond the Proofs: Two More Perspectives

The five proofs above are by no means exhaustive:

- **Inversive Geometry (Morley, 1899):** Morley originally studied variable cardioids tangent to a triangle's extended sides. He discovered that the *centers* of these cardioids precisely track the 27 intersection points of the general angle trisectors.
- **Holonomy (Differential Geometry):** Using marked similarity triangles (MSTs) around an internal vertex, we can calculate a local scale factor called *holonomy*. Proving this holonomy equals 1 guarantees the geometry maps onto a flat plane perfectly, without angle gaps.

Thank You

"Mathematics is the art of giving the same name to different things."
— Henri Poincaré

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